# Module 3: Deal Structuring and the Investment Memo

Days 7, 8 and 9

Module 2 followed a startup from its first data signal through due diligence to a verdict. Module 3 starts at the moment the verdict is "yes" and answers three questions. In what form does the money arrive? What do the terms on the page say? How does the deal team put the case in writing so the investment committee can vote on it?

| Day | Theme | Mandatory topics | Optional topics | Hands-on output |
|---|---|---|---|---|
| 7 | Deal Types and Valuation | 2 | 2 | Valuation and dilution quiz |
| 8 | Term Sheets and Cap Table Lab | 2 | 2 | Two-round cap table and exit waterfall lab |
| 9 | Investment Memo Clinic and Assignment Briefing | 3 | none | Memo template and assignment brief |

Sources are cited in the text as [R#] and listed with links in the References section at the end of the document.

## Day 7: Deal Types and Valuation

A founder and an investor shake hands on "₹2 crore at a ₹10 crore valuation." Two weeks later the founder expects to have sold 16.7% of the company and the investor expects to own 20%. Both are reading the same sentence correctly. One of them assumed ₹10 crore was the pre-money valuation, the other assumed it was post-money. This single ambiguity, and the choice of instrument that carries the money, decide who owns what for the rest of the company's life. Day 7 gives you the vocabulary and the arithmetic to remove that ambiguity.

### Basic (Mandatory)

#### Topic 1: Deal Types — Equity, SAFE, Convertible Notes, and When Each Is Used

Think of the three instruments as three ways to buy a ticket to a future show. Priced equity is buying the ticket today at a printed price. A SAFE is paying today and agreeing that the price will be printed later. A convertible note is the same arrangement bought on credit: the amount owed grows with interest until the price is printed or the bill falls due.

**1. Priced equity (a priced round)**

The investor subscribes to shares, almost always preferred shares, at a fixed price per share. Valuation is therefore agreed on day one. Preferred shares carry rights that common shares do not: a liquidation preference, anti-dilution protection, protective provisions and often a board seat (all covered on Day 8). A priced round is documented through a full package. The NVCA model documents, the US industry baseline, comprise a term sheet, a stock purchase agreement, a charter amendment, an investor rights

agreement, a voting agreement and a right of first refusal and co-sale agreement [R17][R20]. Priced rounds take longer and cost more in legal fees than the two instruments below, which is why they usually begin when a lead investor is ready to set a price.

**2. SAFE (Simple Agreement for Future Equity)**

Y Combinator introduced the SAFE in 2013 and replaced the original pre-money version with the post-money SAFE in September 2018 [R4][R6]. A SAFE is not debt. It carries no interest and no maturity date. The investor pays cash now and receives a contractual right to shares when the company later closes a priced equity financing, at a price that reflects either a valuation cap, a discount, or both [R1][R2].

| SAFE variant | What it sets | Typical use |
|---|---|---|
| Valuation cap, no discount | A ceiling on the price at which the SAFE converts | The default in most rounds, because it fixes the ownership sold: amount ÷ post-money cap [R5] |
| Discount, no cap | A percentage reduction on the next round's price | Rounds where investor and founder cannot agree on any valuation ceiling |
| Uncapped MFN (most favoured nation) | The SAFE takes the best terms of any later SAFE | Very early rounds where the investor wants parity with later SAFE holders [R5] |

YC hosts the current post-money SAFE forms, an optional pro rata side letter, and country variants for Canada, the Cayman Islands and Singapore on its documents page [R1][R5]. If the company is sold or dissolved before the SAFE converts, the investor is paid out ahead of common stockholders, in the amount of the purchase price (or the as-converted value in a sale, if higher) [R1][R2].

**3. Convertible note**

A convertible note is a loan that converts into equity at a later financing. It has the same cap and discount levers as a SAFE and adds three debt features: an interest rate that accrues on the principal, a maturity date, and a repay-or-convert decision when the note matures. Accrued interest usually converts into shares along with the principal, so a note holder ends the period with more shares than a SAFE holder who invested the same cash. Notes are common for bridge rounds and in jurisdictions where a SAFE has no clean legal home.

**4. Which instrument when**

| Situation | Usual instrument | Reason |
|---|---|---|
| Lead investor ready to set a price, board seat wanted | Priced equity | Valuation, rights and control are all agreed in one document |
| Pre-seed, several angels, no lead, speed matters | SAFE (or its Indian equivalent) | Short standard form, low legal cost, no valuation negotiation |

| Situation | Usual instrument | Reason |
|---|---|---|
| Short bridge between two priced rounds | Convertible note or SAFE | Avoids re-pricing the company mid-cycle |
| Investor wants interest and a repayment backstop | Convertible note | Debt features |
| Early cheque below ₹25 lakh into a DPIIT-recognised Indian startup | Priced equity or an iSAFE structured as CCPS | A convertible note needs at least ₹25 lakh per investor in a single tranche [R7][R8] |

**5. The Indian position**

A US-style SAFE has no clean legal home under Indian company law, and a raw US SAFE risks being treated as a deposit under the Companies Act and FEMA [R9]. Indian startups therefore use one of the following [R7][R8][R9]:

- Compulsorily convertible preference shares (CCPS) or compulsorily convertible debentures (CCD), the legal form of the "iSAFE", which converts on a qualifying round and carries no redemption feature.
- A DPIIT convertible note: only a DPIIT-recognised startup may issue it, each investor must put in at least ₹25 lakh in a single tranche, and the note must convert or be repaid within ten years (the limit was earlier five).
- Ordinary priced equity or CCPS in a priced round.

Money taken through a convertible note is excluded from the definition of a "deposit" only if a single investor puts in ₹25 lakh or more in a single tranche [R7]. Foreign investors can buy a convertible note from a DPIIT-recognised startup under Rule 18 of the NDI Rules, while an iSAFE structured as CCPS is treated as pure equity from the date of allotment under FEMA [R7].

**Key takeaways**

- Priced equity sets valuation now. SAFEs and notes defer it.
- A SAFE is a contract for future shares with no interest and no maturity. A note is a loan with both.
- In India the choice is shaped by DPIIT recognition, the ₹25 lakh floor and FEMA, not only by preference.

**Topic 2: Valuation Basics — Pre-Money vs Post-Money, Dilution, Quiz**

**Definitions and formulas**

- Pre-money valuation: what the company is worth before the new cash comes in.
- Post-money valuation = pre-money valuation + new investment.
- Investor ownership = investment ÷ post-money valuation.

- Price per share = pre-money valuation ÷ fully diluted shares before the round.
- New shares issued = investment ÷ price per share.
- Existing holders' dilution factor = 1 − investor's ownership (and a further reduction if an option pool is created in the same round).

**Worked example: Veda Labs seed round**

Veda Labs has two founders holding 14,00,000 shares between them (Founder A 8,40,000, Founder B 5,60,000). A seed fund offers ₹1.75 crore at a ₹7 crore pre-money valuation.

| Step | Calculation | Result |
|---|---|---|
| Post-money valuation | ₹7 cr + ₹1.75 cr | ₹8.75 cr |
| Investor ownership | ₹1.75 cr ÷ ₹8.75 cr | 20% |
| Price per share | ₹7 cr ÷ 14,00,000 | ₹50 |
| New shares to the seed fund | ₹1.75 cr ÷ ₹50 | 3,50,000 |
| Total shares after the round | 14,00,000 + 3,50,000 | 17,50,000 |
| Check | 17,50,000 × ₹50 | ₹8.75 cr (equals post-money) |

Ownership after the round: Founder A 48%, Founder B 32%, seed fund 20%. The founders moved from 100% to 80% combined and the dilution factor was 0.80.

**The ambiguity from the start of the day**

| Reading of "₹1.75 cr at ₹7 cr" | Post-money | Investor ownership |
|---|---|---|
| ₹7 cr is pre-money | ₹8.75 cr | 20.0% |
| ₹7 cr is post-money | ₹7 cr | 25.0% |

The gap is five percentage points of the company, which is why term sheets always state whether the figure is pre-money or post-money and whether it is fully diluted.

**Dilution across rounds**

Dilution compounds because each round multiplies the founders' stake by (1 − new investor percentage).

| Round | New investors sell | Founders' combined ownership |
|---|---|---|
| Start | none | 100.00% |
| Seed | 20% | 80.00% |
| Series A | 20% | 64.00% |

| Round | New investors sell | Founders' combined ownership |
|---|---|---|
| Series B | 15% | 54.40% |
| Series C | 10% | 48.96% |

Owning a smaller percentage of a larger company is the aim. Founders who started at 100% and own 48.96% after Series C are better off if the company's value rose by more than the 2.04x that their stake shrank. Option pools make the reduction steeper than the table shows, which is covered in the advanced topic and on Day 8.

**Quiz: valuation and dilution**

1. A startup has a pre-money valuation of ₹12 crore and raises ₹3 crore. What is the post-money valuation and the investor's ownership?
2. An investor pays ₹5 crore for 25% of a company. What are the post-money and pre-money valuations?
3. Founders hold 70%. A new round sells 20% of the post-round company to new investors. What do the founders hold afterwards?
4. An investor offers ₹2 crore "at a ₹10 crore valuation." What ownership does the investor receive if the figure is pre-money, and if it is post-money?
5. A company has 18,00,000 fully diluted shares and a pre-money valuation of ₹9 crore. It raises ₹2.25 crore. What is the price per share and how many new shares are issued?
6. Name two features that a convertible note has and a SAFE does not.
7. An investor signs a post-money SAFE for ₹60 lakh with a post-money cap of ₹6 crore. What percentage of the company does the SAFE represent before the next round?
8. A DPIIT-recognised startup is offered ₹15 lakh by a resident angel. Can it use a DPIIT convertible note? State the reason.

**Answer key**

| Q | Answer |
|---|---|
| 1 | Post-money ₹15 crore. Investor owns 3 ÷ 15 = 20%. |
| 2 | Post-money = 5 ÷ 0.25 = ₹20 crore. Pre-money = ₹15 crore. |
| 3 | 70% × (1 − 0.20) = 56%. |
| 4 | Pre-money reading: 2 ÷ 12 = 16.67%. Post-money reading: 2 ÷ 10 = 20%. |
| 5 | Price = ₹9 crore ÷ 18,00,000 = ₹50. New shares = ₹2.25 crore ÷ ₹50 = 4,50,000. |
| 6 | Interest that accrues on the principal, and a maturity date at which it must be repaid or converted. |

| Q | Answer |
|---|---|
| 7 | 60 lakh ÷ 6 crore = 10%. |
| 8 | No. A convertible note requires at least ₹25 lakh from a single investor in a single tranche [R8]. Priced equity or an iSAFE structured as CCPS remain available [R8][R9]. |

**Learning pathway for Topics 1 and 2**

- Y Combinator, SAFE financing documents [R1]
- Alexander Jarvis, Post-money SAFE cap explained line by line [R2]
- Warp, Y Combinator SAFE explained [R4]
- Brad Feld and Jason Mendelson, Venture Deals [R40]
- Khurana and Khurana, Convertible notes and SAFE-style instruments in India [R7]

## Advanced (Optional)

**Topic 1: Note and SAFE Mechanics — Caps, Discounts, Conversion Math, Option Pool Effects**

**Cap and discount: the lowest price wins**

A SAFE or note holder converts at the lower of two prices, so the instrument always gives the better of the two protections.

- Cap price = valuation cap ÷ capitalization used in the instrument. In the YC post-money SAFE the "Safe Price" is the post-money valuation cap divided by the Company Capitalization [R2][R3].
- Discount price = next round price × (1 − discount).
- Conversion price = the lower of the cap price and the discount price.
- Shares issued = amount converting ÷ conversion price.

**Worked example: SAFE conversion (Tara Labs)**

Tara Labs' founders hold 10,00,000 shares. An angel signs a post-money SAFE for ₹1 crore with a ₹10 crore post-money cap and a 20% discount. Because the cap is post-money, the SAFE represents 1 ÷ 10 = 10% of the capitalization, so the capitalization including the SAFE is 10,00,000 ÷ 0.90 = 11,11,111 shares and the SAFE converts into 1,11,111 shares. Cap price = ₹10 crore ÷ 11,11,111 = ₹90.

| Next round price | Discount price (20% off) | Cap price | Conversion price | SAFE shares | Value at round price |
|---|---|---|---|---|---|
| ₹180 (strong round) | ₹144 | ₹90 | ₹90 (cap binds) | 1,11,111 | ₹2.00 cr (2.0x) |
| ₹99 (modest round) | ₹79.20 | ₹90 | ₹79.20 (discount binds) | 1,26,263 | ₹1.25 cr (1.25x) |

The cap matters when the company does well and the discount matters when it does modestly. The investor's paper return is at least 1 ÷ (1 − discount) = 1.25x whenever the discount is available.

**Worked example: convertible note conversion**

Instead of a SAFE, suppose Tara Labs issues a ₹50 lakh note with 8% simple interest, a 20% discount and an ₹8 crore pre-money cap, against 10,00,000 pre-money fully diluted shares. Eighteen months later the company closes a Series A at ₹120 per share.

| Step | Calculation | Result |
|---|---|---|
| Accrued interest | ₹50 lakh × 8% × 1.5 years | ₹6 lakh |
| Amount converting | ₹50 lakh + ₹6 lakh | ₹56 lakh |
| Discount price | ₹120 × 0.80 | ₹96 |
| Cap price | ₹8 crore ÷ 10,00,000 | ₹80 |
| Conversion price | lower of ₹96 and ₹80 | ₹80 |
| Shares issued | ₹56 lakh ÷ ₹80 | 70,000 |
| Value at Series A price | 70,000 × ₹120 | ₹84 lakh (1.5x) |

**SAFE stacking**

With post-money SAFEs each holder's percentage is fixed at signing, so every additional SAFE comes out of the founders' stake. Three SAFEs of ₹50 lakh each on a ₹10 crore post-money cap sell 5% each, so the founders are at 85% before any priced round. Warp's guide makes the same point: stacking post-money SAFEs at different caps makes dilution easy to compute per SAFE and harder to see across a whole round [R4].

**Option pool effects**

Investors usually require an unallocated option pool to exist before they invest and count it in the pre-money valuation. The pool therefore dilutes the existing holders and not the new investor, which lowers the effective pre-money valuation. Using the Veda Labs Series A from Day 8: a ₹10 crore investment at a ₹40 crore headline pre-money with a 10% post-money pool.

| Measure | Without pool in pre-money | With 10% post-money pool in pre-money |
|---|---|---|
| Post-money valuation | ₹50 cr | ₹50 cr |
| Pool value (10% of post-money) | not applicable | ₹5 cr |
| Effective pre-money for existing holders | ₹40 cr | ₹35 cr |
| Price per share | ₹228.57 (₹40 cr ÷ 17,50,000) | ₹200 (₹35 cr ÷ 17,50,000) |

The headline pre-money is ₹40 crore but existing shareholders are priced at ₹35 crore, a 12.5% reduction. In the YC post-money SAFE the pool that exists before the round is counted in the capitalization, so a SAFE holder's percentage is protected against it but not against a pool increase made in connection with the priced round [R1][R2]. Founders negotiate the pool size against a hiring plan so the pool covers the next 12 to 18 months of grants and no more.

**Topic 2: Valuation Methods and Indian Instrument Differences**

**Method 1: comparables**

Value the startup against recent financings or acquisitions of similar companies, using a multiple such as enterprise value to annual recurring revenue (EV/ARR) or revenue. Strengths: market-anchored and quick. Limits: early-stage companies often lack revenue, true comparables are rare, and private-market multiples shift with funding cycles. Analysts adjust for stage, growth, margin and geography, and apply a discount for illiquidity.

**Method 2: the VC method**

The VC method works backward from a projected exit to what the company must be worth today for the investor to earn a target return [R12][R15]. The formulas [R12][R13]:

- Post-money = exit value ÷ (1 + target return)^years, or equivalently exit value ÷ target multiple.
- Pre-money = post-money − investment.
- Required final ownership = investment ÷ post-money.
- Required current ownership = required final ownership ÷ retention ratio, where retention ratio = product of (1 − dilution) across future rounds.

Worked example. A fund invests ₹5 crore and expects an exit in 6 years at 4x projected revenue of ₹100 crore, so exit value is ₹400 crore. The fund targets a 10x multiple, which is an annual return of 10^(1/6) − 1 = 46.4%. It expects two future rounds of 20% dilution each.

| Step | Calculation | Result |
|---|---|---|
| Exit value | ₹100 cr × 4 | ₹400 cr |
| Post-money, ignoring dilution | ₹400 cr ÷ 10 | ₹40 cr |
| Required final ownership | ₹5 cr ÷ ₹40 cr | 12.5% |
| Retention ratio | 0.80 × 0.80 | 0.64 |
| Required current ownership | 12.5% ÷ 0.64 | 19.53% |
| Post-money today | ₹5 cr ÷ 0.1953 | ₹25.6 cr |
| Pre-money today | ₹25.6 cr − ₹5 cr | ₹20.6 cr |

The method is sensitive to the exit value and the target multiple, and Equidam's treatment shows target multiples for early-stage funds are commonly around 5x on the fund and much higher on a single

successful company, because most investments fail [R14]. The VC method fits early-stage deals where a discounted cash flow has nothing reliable to discount; Damodaran-style comparisons of the two show where each breaks down [R16].

**Indian instrument differences**

Indian instrument rules change by notification, so confirm the current position with your VC partner before applying any of the following to a live deal.

| Instrument | Who can issue | Floor or condition | FEMA position for foreign investors | Notes |
|---|---|---|---|---|
| Equity shares | Any company | None | Equity instrument, subject to pricing guidelines | Valuation fixed now |
| CCPS | Any company | Must convert compulsorily | Treated as equity, subject to pricing floor | The legal form of the iSAFE [R7][R9] |
| CCD | Any company | Must convert compulsorily | Equity instrument if fully and compulsorily convertible | Used where a debenture form is preferred |
| DPIIT convertible note | DPIIT-recognised startup | ₹25 lakh per investor per tranche; convert or repay within 10 years | Permitted for non-residents under Rule 18 of the NDI Rules | Not a "deposit" at the ₹25 lakh floor [R7][R8] |
| US-style SAFE | Not recognised as such in Indian company law | Not applicable | Risk of recharacterisation as a deposit | Raise at a Delaware parent, then capitalise the Indian entity through permitted instruments [R9][R10] |

Three further points shape valuation in India:

- Pricing floor: for a foreign investor, the FEMA pricing requirement limits how far a cap or discount can reduce the conversion price below fair value at the time the note was issued [R8].
- Valuation postponed, not removed: an iSAFE delays the price discussion but the issue of CCPS still needs a valuation under the Companies Act and FEMA [R7].
- Angel tax: Section 56(2)(viib) of the Income Tax Act, which taxed share premium above fair market value, was abolished by the Finance (No. 2) Act, 2024 and does not apply from 1 April 2025, for resident and non-resident investors alike. Share premium received earlier can still be assessed [R38][R39].

**Learning pathway for Advanced topics**

- Ryan O'Connell, The VC method of valuation [R12]
- Equidam, Understanding the venture capital method [R14]
- AnalystNotes (CFA curriculum), A note on valuation of venture capital deals [R15]
- TaxTMI, Convertible note compliance for foreign investors [R8]
- Legal500, Abolition of angel tax in India [R38]

## Day 8: Term Sheets and Cap Table Lab

A term sheet runs two or three pages and quietly settles who is paid first in a sale, who sits on the board, who can veto a decision and how a bad year is shared. Valuation draws the attention, but in a downside outcome the liquidation preference decides more money than the headline price. Day 8 teaches you to read the document clause by clause and then to read the cap table that results from it.

### Basic (Mandatory)

#### Topic 1: Key Term-Sheet Terms in Plain Language

A term sheet is a summary of the principal terms of an investment. It is non-binding except for provisions such as confidentiality and exclusivity, and the lawyers convert it into definitive documents. In the US most term sheets start from the NVCA model, whose annotations explain each provision [R17][R18]. In Indian deals the term sheet leads to a share subscription agreement and a shareholders' agreement, and preference rights are built into the terms of the CCPS [R24].

| Term | Plain-language meaning | Market position | What to check |
|---|---|---|---|
| Valuation | The price of the company and the basis it is quoted on | Stated pre-money, fully diluted | Pre- or post-money, and whether the option pool sits inside the pre-money |
| Investment amount and price per share | Cash in, and the price each new share costs | Price = pre-money ÷ fully diluted shares | Share count used in the denominator |
| Liquidation preference | Who is paid first, and how much, when the company is sold or wound up | 1x. Carta deal data reported by Vaquill puts about 98% of deals at 1x and about 95% at non-participating in Q2 2025 [R21] | Multiples above 1x, participation, cumulative dividends [R19][R20] |
| Participation | Whether the investor takes the preference and then also shares in what is left (the "double dip") | Non-participating | Participating without a cap [R22][R25] |

| Term | Plain-language meaning | Market position | What to check |
|---|---|---|---|
| Dividends | Payments to preferred holders | Non-cumulative, only if the board declares them | Cumulative dividends grow the preference over time [R20] |
| Anti-dilution | Adjusts the investor's conversion price if a later round is cheaper (a down round) | Broad-based weighted average [R17][R27] | Full ratchet, narrow base, missing carve-outs [R19] |
| Pro-rata right | The right to buy a share of future rounds to keep one's percentage | Granted to lead and significant investors | Who holds it, and whether it expires |
| Board seat | How many directors each side appoints | Three to five seats at seed and Series A, scaled to deal size and ownership [R20] | Who holds the tie-break |
| Protective provisions | Actions that need the investor's consent: new senior shares, sale of the company, charter changes, large debt | Standard list in the NVCA form [R17] | Thresholds, and whether the list stretches into day-to-day management |
| Vesting | Founders earn their shares over time | Four years with a one-year cliff | Credit for time already served, acceleration on a sale |
| Option pool | Shares reserved for future hires | Sized to the next 12 to 18 months of hiring | Counted in the pre-money (see Day 7) |
| Drag-along, ROFR, co-sale | Forces minority holders into a sale, gives the company or investors first call on transfers, lets investors join a founder's sale | In the NVCA investor documents [R20] | Drag-along thresholds |
| Exclusivity | The company stops shopping the deal for a stated period | 30 to 60 days | Length, and what lets the clock stop |

**One example for each of three terms**

- Pro-rata: Series A holds 20%. In a ₹30 crore Series B it buys 20% of the new shares, which is ₹6 crore, to keep its percentage.
- Vesting: a founder with 8,40,000 shares on four-year vesting with a one-year cliff has nothing vested at month 9, 2,10,000 shares (25%) at month 12, and then 17,500 shares a month for 36 months. At month 30 the vested total is 2,10,000 + 18 × 17,500 = 5,25,000.

- Liquidation preference: Series A invests ₹10 crore with a 1x non-participating preference. In a ₹30 crore sale its 20% would be worth ₹6 crore as common, so it takes the ₹10 crore preference instead.

**Reading exercise: Veda Labs Series A, summary of terms (excerpt)**

- Investment: ₹10 crore in Series A CCPS.
- Valuation: ₹40 crore pre-money, fully diluted, including an unallocated option pool equal to 10% of the post-money capitalization.
- Liquidation preference: 1x, non-participating.
- Anti-dilution: broad-based weighted average.
- Pro-rata: Series A may participate in later rounds to maintain its percentage.
- Board: three seats, two nominated by the founders and one by Series A.
- Vesting: founder shares vest over four years with a one-year cliff.
- Exclusivity: 45 days.

Questions: (1) What are the post-money valuation and the Series A percentage? (2) What is the effective pre-money valuation for existing holders? (3) What does Series A receive in a ₹30 crore sale? (4) Is any clause more aggressive than the market position in the table above?

Answers: (1) Post-money ₹50 crore and Series A owns 10 ÷ 50 = 20%. (2) The pool is 10% of ₹50 crore = ₹5 crore, so existing holders are priced at ₹35 crore. (3) ₹10 crore, the preference, because converting would give 20% × ₹30 crore = ₹6 crore. (4) No. Every term sits at the market position, and the one to watch is the pool, which lowers the effective price by 12.5%.

**Key takeaways**

- Valuation sets the price and the other terms set what the price is worth in each outcome.
- Three terms carry most of the economics after valuation: the multiple and participation of the liquidation preference, the type of anti-dilution, and the option pool.
- Control sits in the board composition and the protective provisions.

**Topic 2: Reading a Simple Cap Table Across Two Rounds**

A capitalization table lists every holder of the company's securities, how many shares each holds and what percentage that represents on a fully diluted basis. Read it as a ledger with three jobs: it shows who owns what today, it proves each round's arithmetic, and it is the base for every later exit calculation.

**Veda Labs after founding, the Seed round and the Series A**

| Holder | At founding | After Seed | After Series A |
|---|---|---|---|
| Founder A | 8,40,000 (60.0%) | 8,40,000 (48.0%) | 8,40,000 (33.6%) |

| Holder | At founding | After Seed | After Series A |
|---|---|---|---|
| Founder B | 5,60,000 (40.0%) | 5,60,000 (32.0%) | 5,60,000 (22.4%) |
| Seed fund | none | 3,50,000 (20.0%) | 3,50,000 (14.0%) |
| Option pool | none | none | 2,50,000 (10.0%) |
| Series A fund | none | none | 5,00,000 (20.0%) |
| **Total shares** | **14,00,000** | **17,50,000** | **25,00,000** |

| Round | Price per share | New shares | Money raised | Post-money valuation | New investor stake |
|---|---|---|---|---|---|
| Seed | ₹50 | 3,50,000 | ₹1.75 cr | ₹8.75 cr | 20% |
| Series A | ₹200 | 5,00,000 to the fund, 2,50,000 to the pool | ₹10 cr | ₹50 cr | 20% |

How the Series A column is derived: Series A sells 20% and the pool is 10% of the post-money company, so the 17,50,000 existing shares are the remaining 70%. Total shares are 17,50,000 ÷ 0.70 = 25,00,000, the fund receives 20% × 25,00,000 = 5,00,000 shares and the pool receives 2,50,000. The price is ₹50 crore ÷ 25,00,000 = ₹200.

**Six checks to run on any cap table**

1. Shares in the total row equal the sum of the rows above it.
2. Percentages sum to 100%.
3. Total shares × the round price equals the post-money valuation (25,00,000 × ₹200 = ₹50 crore).
4. New shares × the round price equals the money raised (5,00,000 × ₹200 = ₹10 crore).
5. Existing holders' share counts did not change between rounds. Only their percentages did.
6. The pool is treated as the term sheet says: here inside the pre-money.

**Guided questions**

| Question | Answer |
|---|---|
| How far has Founder A been diluted since founding? | From 60% to 33.6%, a factor of 0.56 (0.80 × 0.70) |
| What is the seed fund's paper value after the Series A? | 3,50,000 × ₹200 = ₹7 crore on ₹1.75 crore invested, 4.0x |

| Question | Answer |
|---|---|
| What is the effective pre-money valuation of the Series A? | 17,50,000 existing shares × ₹200 = ₹35 crore against a ₹40 crore headline |
| The seed fund sold nothing. Why did it fall from 20% to 14%? | Its stake was multiplied by (1 − 0.20 − 0.10) = 0.70, because the Series A and the pool both came out of the existing holders |
| How much must the Series A fund invest in a ₹30 crore Series B to hold its 20%? | 20% × ₹30 crore = ₹6 crore |

**Learning pathway for Topics 1 and 2**

- NVCA, Model term sheet [R17]
- MIT Orbit, Term sheet pamphlet for founders [R18]
- VCBeast, VC term sheet template and guide [R19]
- VCBeast, NVCA term sheet template [R20]
- Brad Feld and Jason Mendelson, Venture Deals [R40]

## Advanced (Optional)

**Topic 1: Build a Cap Table from Scratch and Run an Exit Waterfall**

**Building the cap table in a spreadsheet**

Use one input block and one row per holder.

| Input | Symbol | Veda Labs Series A |
|---|---|---|
| Existing fully diluted shares | E | 17,50,000 |
| New investor stake in the post-money company | a | 20% |
| Option pool as a share of the post-money company | p | 10% |
| Post-money valuation | P | ₹50 crore |

Formulas: total post-round shares N = E ÷ (1 − a − p); new investor shares = a × N; pool shares = p × N; price per share = P ÷ N. In a sheet this is a single cell, `=E/(1-a-p)`, with the others referencing it. Add the six checks from Topic 2 as visible cells that return TRUE or FALSE.

**Waterfall rules**

Proceeds of a sale are paid in this order: debt and transaction costs, then preferred shares by seniority (pari passu holders share pro rata within a level), then common stock and any preferred that converted. Each preferred class then follows its own rule [R22][R23][R25][R26]:

- Non-participating: the holder takes the greater of its preference and its as-converted share. It never takes both.
- Participating: the holder takes its preference and then also shares in the remainder as if converted.
- Capped participating: as above, but total proceeds to the holder are limited to a multiple of the investment, and the holder converts to common if that is worth more.

Shoosmiths' guidance to founders is to ask the investor to model the waterfall at several exit values before signing [R23]. Do that in a table with exit value down the side and holders across the top.

**Veda Labs waterfall**

Assumptions: Series A is senior with a 1x preference; the seed fund holds a 1x non-participating preference and converts when that pays more; the pool is fully granted with a negligible exercise price; there is no debt and no transaction cost. Figures are in ₹ crore.

Series A 1x non-participating:

| Exit value | Founders | Seed fund | Option pool | Series A |
|---|---|---|---|---|
| 10 | 0.00 | 0.00 | 0.00 | 10.00 |
| 30 | 14.00 | 3.50 | 2.50 | 10.00 |
| 60 | 33.60 | 8.40 | 6.00 | 12.00 |
| 150 | 84.00 | 21.00 | 15.00 | 30.00 |

Series A 1x participating (uncapped):

| Exit value | Founders | Seed fund | Option pool | Series A |
|---|---|---|---|---|
| 10 | 0.00 | 0.00 | 0.00 | 10.00 |
| 30 | 11.20 | 2.80 | 2.00 | 14.00 |
| 60 | 28.00 | 7.00 | 5.00 | 20.00 |
| 150 | 78.40 | 19.60 | 14.00 | 38.00 |

How the ₹30 crore row is built. Non-participating: Series A converts only if 20% × 30 = ₹6 crore beats ₹10 crore, so it takes ₹10 crore. The remaining ₹20 crore is shared by the 20,00,000 other shares, and the seed fund converts (17.5% × 20 = ₹3.5 crore beats its ₹1.75 crore preference). Participating: Series A takes ₹10 crore, then 20% of the remaining ₹20 crore, which is ₹4 crore, for ₹14 crore in total.

What the table says: at a ₹30 crore sale Series A holds 20% of the company and receives 33% of the proceeds if non-participating and 47% if participating. Founders receive ₹2.8 crore less under participation at ₹30 crore and ₹5.6 crore less at ₹60 crore and ₹150 crore. The gap equals 20% of the ₹10 crore preference, because the preference is deducted before the remainder is shared.

Breakpoints worth knowing:

- A non-participating Series A converts when 20% of the exit exceeds ₹10 crore, so above ₹50 crore.
- With a 3x cap on participation (₹30 crore total), Series A reaches the cap at an exit of ₹110 crore (10 + 0.20 × (exit − 10) = 30) and stays at ₹30 crore until it is better off converting, above ₹150 crore.

**Cap table lab: Rhea Labs**

Rhea Labs' founders hold 18,00,000 shares. A seed fund invests ₹2.25 crore at a ₹9 crore pre-money valuation. Later, a Series A fund invests ₹12 crore for 20% of the post-money company, and a 5% post-money option pool is created inside the pre-money valuation.

Tasks: (a) build the post-Seed table; (b) build the post-Series A table and find the price per share; (c) find each holder's percentage and the effective pre-money valuation; (d) run the waterfall at exit values of ₹40 crore and ₹100 crore with the Series A on a 1x non-participating and then a 1x participating preference (Series A senior; seed 1x non-participating, converting when better; pool fully granted).

Answer key:

| Item | Result |
|---|---|
| (a) Post-Seed | Price ₹50; 4,50,000 new shares; 22,50,000 total; founders 80%, seed 20%; post-money ₹11.25 crore |
| (b) Post-Series A | Existing 22,50,000 shares are 75%, so N = 30,00,000. Series A 6,00,000 shares, pool 1,50,000 shares, price ₹200, post-money ₹60 crore, headline pre-money ₹48 crore |
| (c) Ownership | Founders 60%, seed fund 15%, pool 5%, Series A 20%. Effective pre-money = 22,50,000 × ₹200 = ₹45 crore |
| (d) Breakpoint | A non-participating Series A converts above ₹12 crore ÷ 0.20 = ₹60 crore |

| Waterfall (₹ crore) | Founders | Seed fund | Pool | Series A |
|---|---|---|---|---|
| Exit 40, non-participating | 21.00 | 5.25 | 1.75 | 12.00 |
| Exit 40, participating | 16.80 | 4.20 | 1.40 | 17.60 |
| Exit 100, non-participating | 60.00 | 15.00 | 5.00 | 20.00 |
| Exit 100, participating | 52.80 | 13.20 | 4.40 | 29.60 |

**Topic 2: Anti-Dilution Types and Negotiation Dynamics**

Anti-dilution applies only when shares are issued below the protected series' current conversion price, and it protects against price, not against ordinary dilution [R27]. The investor does not receive new shares. The conversion ratio of the preferred into common changes instead [R30].

**The two formulas**

- Full ratchet: the new conversion price equals the price of the new issue, whatever its size [R27][R29].
- Broad-based weighted average: CP2 = CP1 × (A + B) ÷ (A + C), where CP1 is the conversion price before the issue, A is the fully diluted shares outstanding before it, B is the new money divided by CP1 (the shares that money would have bought at the old price) and C is the shares actually issued [R27][R28]. Narrow-based weighted average uses the same formula with a smaller A, which produces a larger adjustment [R28].

**Worked example: Veda Labs down round**

Series A holds 5,00,000 shares converting 1:1 at CP1 = ₹200, with 25,00,000 fully diluted shares outstanding. A Series B raises ₹5 crore at ₹100 per share, which issues 5,00,000 new shares (C).

| Quantity | Broad-based weighted average | Full ratchet |
|---|---|---|
| B | ₹5 cr ÷ ₹200 = 2,50,000 | not used |
| New conversion price CP2 | 200 × (25,00,000 + 2,50,000) ÷ (25,00,000 + 5,00,000) = ₹183.33 | ₹100 |
| Conversion ratio | 200 ÷ 183.33 = 1.0909 | 2.0 |
| Series A as-converted shares | 5,45,455 | 10,00,000 |
| Value of those shares at ₹100 | ₹5.45 cr | ₹10.00 cr |

Ownership after Series B (the seed fund holds no anti-dilution right):

| Holder | No protection | Broad-based weighted average | Full ratchet |
|---|---|---|---|
| Founder A | 28.00% | 27.58% | 24.00% |
| Founder B | 18.67% | 18.39% | 16.00% |
| Seed fund | 11.67% | 11.49% | 10.00% |
| Option pool | 8.33% | 8.21% | 7.14% |
| Series A | 16.67% | 17.91% | 28.57% |
| Series B | 16.67% | 16.42% | 14.29% |

Full ratchet restores the Series A's ₹10 crore of value in full, and it moves Founder A from 28.00% to 24.00% to do it. Weighted average moves Founder A by 0.42 points. A ₹5 crore round at half the price is a bigger event than the ratchet's wording suggests, and the formula never considers how small the new round is. Weighted average does consider it, which is why it is the market standard [R27][R29].

**Negotiation dynamics**

| Lever | Founder's position | Investor's position |
|---|---|---|
| Type | Broad-based weighted average | Full ratchet in distressed or high-risk deals |
| Base | Broad base (counts options, warrants, pool) for a smaller adjustment | Narrow base for a larger one [R28] |
| Carve-outs | Option grants, conversions of existing securities and equipment financing do not trigger it [R19] | Fewer carve-outs |
| Pay-to-play | Investors lose protection if they skip the down round | Prefers to keep protection whatever it does |
| Waiver | A majority of the preferred can waive for everyone [R27] | Wants a higher threshold |
| Duration | Falls away at a qualified financing or IPO | Continues until conversion |

Practitioner sources describe broad-based weighted average as the common form in US venture financings, and the NVCA model charter offers it and full ratchet as alternative clauses [R27]. One estimate puts broad-based weighted average at more than 90% of venture deals [R29]. A harsher term is not automatically the better outcome for the investor: Qapita notes that the strongest anti-dilution formula does not guarantee the strongest investment result, since a ratchet that crushes the founders removes their incentive and complicates the next round [R28]. Leverage comes from the competitiveness of the round. A founder with more than one term sheet can usually settle on broad-based weighted average, and an investor who is the only source of capital in a difficult round can ask for more.

**Learning pathway for Advanced topics**

- Altss, Anti-dilution protection [R27]
- Qapita, Anti-dilution provisions [R28]
- Ryan O'Connell, Antidilution provisions: full ratchet vs weighted average [R29]
- Bird & Bird, How does a liquidation preference work [R22]
- Shoosmiths, Liquidation preferences [R23]
- Mondaq India, Liquidation preference: a downside protection [R24]
- Allied VC, Modelling liquidation preferences in cap tables [R26]

## Day 9: Investment Memo Clinic and Assignment Briefing

Every investment starts with a memo. A partner reads the executive summary first, then jumps to the risks, then to the recommendation [R33]. That reading order tells you where to spend the effort: the memo's opening, its risk section and its last paragraph carry the decision. Days 4 to 8 gave you the

inputs: market, team, traction, unit economics, valuation and terms. Day 9 turns them into the one document an investment committee actually votes on, and then hands you the assignment that tests it.

### Topic 1: Memo Structure and One-Page Template

An investment memo is the written argument for why a fund should invest in, keep researching or pass on a startup. Funds call it a deal memo, diligence memo, IC memo or recommendation memo [R35]. Two versions exist in practice [R34]:

| Version | Written when | Purpose |
|---|---|---|
| Deal memo (initial memo) | After two or three conversations with the founder | Decide whether the opportunity deserves full diligence |
| Formal IC memo | After full diligence is complete | Secure the committee's approval to deploy capital |

A survey of 54 investment organisations found that 97% used investment memos and 86% used standard templates [R36]. For a seed cheque of roughly $500,000 to $2 million, expect a memo of three to five pages [R33].

**The twelve sections**

| # | Section | Question it answers | Draws on |
|---|---|---|---|
| 1 | Recommendation and ask | Invest, pass or watch? How much, on what terms? | Day 7 and Day 8 |
| 2 | Company snapshot | What does it do, at what stage, in what round? | Module 2, Day 4 |
| 3 | Investment thesis | Why does this company win, and why now? | Sequoia's "why now" [R31] |
| 4 | Problem and solution | Whose pain is it, and how is it solved today? | Sequoia's problem and solution [R31] |
| 5 | Market | TAM, SAM, SOM, growth, structure | Module 2, Day 4 |
| 6 | Product and moat | What is hard to copy? | Module 2, moat types |
| 7 | Team | Does founder-market fit exist? | Module 2, founder diligence |
| 8 | Traction and unit economics | Do customers stay, and does each one make money? | CAC, LTV, LTV/CAC of at least 3 |
| 9 | Competition | Who else is solving it, and what is the plan to win? | Sequoia's competition [R31] |

| # | Section | Question it answers | Draws on |
|---|---|---|---|
| 10 | Deal terms and returns | Instrument, valuation basis, ownership, preference, board, and what exit makes the fund's money back | Day 7 VC method, Day 8 terms |
| 11 | Risks and mitigants | What could break the thesis? | Module 2, due diligence |
| 12 | Next steps | What is still open in diligence? | Checklist from Module 2 |

Sequoia's guide for founders lists the company purpose, problem, solution, why now, market potential, competition, business model, team, financials and vision [R31]. The memo is the investor-side mirror of that outline, with the investor's judgement added.

**One-page template**

Use this skeleton for a deal memo, then expand each block for the full memo. Target about 500 words.

- Header: company, sector, stage and round, memo date, author.
- Recommendation (two lines): invest, pass or watch. State the amount, the valuation (pre-money and post-money), the instrument and the ownership acquired.
- Thesis (three sentences): why this company wins, why now, and the single metric that best supports it.
- Business (one paragraph): customer, product, how it makes money.
- Evidence table: one row each for team, market, traction, unit economics and moat, with one metric and a one-line reading per row.
- Returns case: the exit value the fund needs. Use the VC method from Day 7: required exit value = investment ÷ (ownership at exit × 1 ÷ target multiple).
- Terms: instrument, valuation basis, liquidation preference, anti-dilution, board seat, pro-rata right.
- Top risks (three to five): each with a mitigant and a decision trigger, which is the evidence that would change the recommendation [R36].
- Open items and next step: the diligence still to do and who does it.

**Key takeaways**

- Write the recommendation first and change it if the evidence changes [R36].
- Length follows cheque size. The structure does not change.
- The risk section and the recommendation get read most closely, so they must be the strongest parts [R33].

## Topic 2: Strong vs Weak Sample Memo, and Writing the Thesis and Risk Sections

The sample company is MedAI Diagnostics from the Module 2 due-diligence case: a seed-stage AI tool that detects lung infections from X-rays, founded by an ex-Google Brain AI researcher and a radiologist with 12 years of clinical experience [R41]. The deal terms in the memo below are illustrative.

**The weak version**

> *MedAI is an exciting healthtech startup using AI to transform diagnostics. The team is strong and the market is huge. They have good traction with hospitals and are growing fast. There are some regulatory risks but we think they can manage them. We recommend considering an investment.*

| Statement | What is wrong | What a strong memo does |
|---|---|---|
| "Exciting" and "transform" | Adjectives in place of evidence | Replaces them with a number or fact |
| "The team is strong" | No reason given | Names the specific fit: model builder plus 12-year clinician |
| "The market is huge" | No size, no source | States the TAM and its source, and the SAM the company can reach |
| "Good traction" and "growing fast" | No metric | States revenue, retention and customer count |
| "Some regulatory risks" | Vague, no mitigant | Names the risk, ranks it, adds a mitigant and a trigger |
| "Consider an investment" | Not a recommendation | States invest, pass or watch, with amount and terms [R33] |

**The strong version (one page)**

**MedAI Diagnostics — Seed — Investment memo (illustrative terms)**

Recommendation. Invest ₹4 crore in a seed CCPS round at ₹16 crore pre-money (₹20 crore post-money), for 20% of the company. Terms: 1x non-participating preference, broad-based weighted-average anti-dilution, one board seat. Closing condition: an independent accuracy-validation protocol agreed with two of the four pilot hospitals.

Thesis. MedAI wins radiology triage because its founders hold both the model and the clinical credibility to get it deployed, and 92% retention across paying hospitals shows the product stays once installed. The market is large: the Module 2 case sizes global diagnostic AI at about $25 billion TAM [R41]. The economics support scale: a customer costs ₹6,000 to acquire and returns ₹48,000, an LTV/CAC of 8:1 against the 3:1 benchmark.

Evidence.

| Area | Metric | Reading |
|---|---|---|
| Team | Ex-Google Brain researcher and a radiologist with 12 years' practice | Covers model and clinic |
| Traction | 4 pilot hospitals, 2 paying, ₹9 lakh monthly revenue | Early but paid |
| Retention | 92% | Strong for B2B |
| Unit economics | Gross margin 82%, CAC ₹6,000, LTV ₹48,000, payback about 1.5 months | Healthy |
| Moat | Proprietary model, clinical relationships, regulatory approval as a barrier | Defensible if approval lands |

Returns case. At 20% today and two future rounds of 20% dilution, the fund's retention ratio is 0.80 × 0.80 = 0.64, so it holds 12.8% at exit. A 10x return on ₹4 crore needs a ₹40 crore stake, which means an exit of ₹40 crore ÷ 0.128 = ₹312.5 crore. At 6x revenue that is ₹52 crore of annual revenue, against ₹1.08 crore annualised today, so the plan needs roughly 48x revenue growth.

Risks, ranked.

| Rank | Risk | Mitigant | Decision trigger |
|---|---|---|---|
| 1 | Regulatory approval is pending, which limits commercial rollout | Roadmap filed; closing condition tied to validation | No filing timeline within the diligence period |
| 2 | The 91% accuracy claim is internal and not externally validated | Independent validation at two pilot hospitals | External accuracy materially below 91% |
| 3 | Revenue rests on two paying hospitals, so losing one removes about half of it | Convert two more pilots to paid | No further conversions in the next two quarters |
| 4 | Hospital sales cycles are long, so growth may lag the returns case | Pilot-to-paid conversion tracked monthly | Conversion time exceeds the plan |

Next steps. Complete reference calls with the four pilot hospitals, review the regulatory roadmap with counsel, and verify the cap table and IP assignments.

**Writing the thesis**

The thesis is the spine of the memo [R35]. Use this formula: [company] wins [market] because [specific edge], made possible by [why now], and the evidence is [metric]. Strong theses are specific enough that a reader could disagree with them, and the best ones are contrarian [R34].

| Weak thesis | Strong thesis |
|---|---|
| Large market, strong team, good traction | MedAI wins radiology triage because its founders hold both the model and the clinical credibility to |

| Weak thesis | Strong thesis |
|---|---|
| | deploy it, and 92% retention across paying hospitals shows it sticks |
| An AI company in a hot sector | The company turns a manual weekly task into a daily one at one-fifth the cost, and three customers have already switched |

**Writing the risk section**

List three to six risks, ranked by severity and likelihood, each with a mitigant and a decision trigger [R33][R36][R37]. The purpose is not to dismiss risks. A memo that shows no risks signals weak analysis, and venture investors back companies despite risks, not in their absence [R37]. Typical categories are market, competition, execution, product, regulatory and financing [R34].

| Weak risk | Strong risk |
|---|---|
| Competition may increase | Two funded incumbents could bundle this feature into their existing contracts within 12 months, which would cap pricing |
| Regulatory risk | Approval is pending and rollout cannot scale without it; we require a filing timeline before closing |
| Execution risk | Both founders are technical and the first sales hire is not yet in place |

**Exercise 1.** Rewrite the weak MedAI paragraph using the data above, ending with a recommendation that states an amount and terms.

**Exercise 2: a memo that says pass.** Use the QuickBasket data from the Module 2 case: 80,000 downloads, ₹22 lakh monthly revenue, 23% retention, CAC ₹450, LTV ₹700, negative contribution margin, payback of five to six months, heavy discounts driving usage [R41]. Write the recommendation, a two-sentence thesis for passing, and the one piece of evidence that would change your mind.

Model answer outline. Recommendation: pass. Reason: LTV/CAC is 700 ÷ 450 = 1.56:1 against the 3:1 benchmark, retention of 23% means growth is bought with discounts, and the company has no moat against better-funded rivals. What would change the view: three consecutive months of retention above 40% with discounts withdrawn and a positive contribution margin.

**Learning pathway for Topics 1 and 2**

- Sequoia Capital, Writing a business plan [R31]
- Visible.vc, Investment memo template [R32]
- VCBeast, How to write an investment memo: VC template and examples [R33]
- VCBeast, Investment committee memo structure [R34]
- Venture Capital Careers, Venture capital investment memo [R35]

- Hustle Fund, Investment memo template [R36]
- VCBeast, VC case study guide [R37]

### Topic 3: Assignment Briefing with the VC Firm's Problem Statement

**The assignment**

Each fellow writes an investment memo and a term-sheet recommendation on the startup assigned by the partner VC firm, then defends it in a short committee-style Q&A. The firm issues its problem statement at the briefing and the frame below is what that statement plugs into.

| Field | Entered at the briefing |
|---|---|
| VC firm and its investment thesis | Firm to complete |
| Company under review and sector | Firm to complete |
| Round, ask and proposed instrument | Firm to complete |
| Materials provided (deck, data room extract, financials) | Firm to complete |
| The question the firm wants answered | Firm to complete |
| Deadline and submission channel | Firm to complete |

**Deliverables**

1. Investment memo of three to five pages in the Topic 1 structure [R33].
2. One-page valuation and dilution sheet: post-money and pre-money valuation, ownership acquired, a before and after cap table (Day 8), and the VC-method returns case (Day 7).
3. Term-sheet recommendation: instrument, valuation basis, and a stated position on liquidation preference, anti-dilution, pro-rata, board and vesting, each with a one-line reason.
4. A five-minute oral recommendation followed by questions from the panel.

**Work plan** (about six and a half hours for a three-page memo; a VC case-study guide suggests four to six hours for a first memo [R37])

| Block | Time | Output |
|---|---|---|
| Read the materials and mark every claim as verified or stated by the company | 30 min | Claims log |
| Market and competition | 60 min | TAM, SAM, SOM and competitor table |
| Team and traction | 60 min | Evidence table rows |
| Unit economics and valuation | 60 min | CAC, LTV, returns case |
| Terms and cap table | 60 min | Term-sheet positions and cap table |

| Block | Time | Output |
|---|---|---|
| Drafting | 90 min | Full memo |
| Risks, recommendation and edit | 30 min | Ranked risks, final recommendation |

**Evaluation rubric**

| Criterion | Weight | What a top submission shows |
|---|---|---|
| Recommendation clarity | 15 | Invest, pass or watch stated in the first lines, with amount and terms |
| Thesis specificity | 20 | A claim an investor could disagree with, supported by one metric |
| Evidence and numbers | 20 | Market, traction and unit economics quantified, with sources |
| Valuation and terms reasoning | 20 | Correct pre- and post-money arithmetic, a cap table, and reasoned term positions |
| Risk quality | 15 | Three to six ranked risks, each with a mitigant and a decision trigger |
| Writing and structure | 10 | Template headings, no filler, every external fact cited |

**Pre-submission checklist**

- The first two lines state the recommendation, the amount and the valuation basis.
- Post-money, pre-money, ownership and price per share reconcile to each other.
- The cap table totals 100% and total shares × price equals post-money.
- Every figure is marked as company-reported or independently verified.
- Each risk has a mitigant and a decision trigger.
- Every external statistic or claim carries a citation with a link.

## References

Citations in the text use these reference numbers.


| Ref | Source | Link |
|---|---|---|
| R1 | Y Combinator, SAFE financing documents | https://www.ycombinator.com/documents |
| R2 | Alexander Jarvis, Post Money SAFE Cap Explained Line by Line | https://www.alexanderjarvis.com/post-money-safe-cap-explained-line-by-line/ |

| Ref | Source | Link |
|---|---|---|
| R3 | Alexander Jarvis, Post Money SAFE Cap and Discount Explained Line by Line | https://www.alexanderjarvis.com/post-money-safe-cap-and-discount-explained-line-by-line/ |
| R4 | Warp, Y Combinator SAFE Explained | https://www.warp.co/blog/y-combinator-safe-explained |
| R5 | VCBeast, How to Fill Out a SAFE | https://vcbeast.com/how-to-fill-out-a-safe |
| R6 | Lowenstein Sandler, Potentially improved tax treatment of the new post-money SAFE | https://www.lowenstein.com/news-insights/client-alerts/potentially-improved-tax-treatment-of-the-new-post-money-safe-tax |
| R7 | Khurana and Khurana, Convertible Notes and SAFE-Style Instruments in India | https://www.khuranaandkhurana.com/convertible-notes-and-safe-style-instruments-in-india |
| R8 | TaxTMI, Convertible note compliance for foreign investors | https://www.taxtmi.com/article/detailed?id=16812 |
| R9 | eChai Ventures, summary of Madhavan Srivatsan (Bar & Bench), Convertible Notes and SAFE Notes in India: The Dilemma | https://echai.ventures/startingup/valuation-safes/can-i-even-use-a-safe-in-india-or-do-i-have-to-do-a-ccps-or-convertible-note |
| R10 | Business Setup, SAFE notes in India (2025) | https://businesssetup.in/safe-notes-in-india/ |
| R11 | IncorpX, Convertible Notes vs SAFE Notes: Which Is Better for Indian Startups | https://www.incorpx.io/blog/convertible-notes-vs-safe-notes-india |
| R12 | Ryan O'Connell, The VC Method of Valuation | https://ryanoconnellfinance.com/venture-capital-valuation-method/ |
| R13 | Equidam, VC Method startup valuation | https://www.equidam.com/vc-method-startup-valuation/ |
| R14 | Equidam, Understanding the Venture Capital Method (paper) | https://www.equidam.com/resources/equidam-vc-method-paper.pdf |
| R15 | AnalystNotes, A note on valuation of venture capital deals (CFA curriculum) | https://analystnotes.com/cfa-study-notes-a-note-on-valuation-of-venture-capital-deals.html |
| R16 | NOVA SBE, Entrepreneurial Finance and Venture Capital, lecture 5 | https://moodle.novasbe.pt/pluginfile.php/672198/mod_resource/content/8/lecture5moodle.pdf |
| R17 | NVCA, Model term sheet | https://nvca.org/recommends/nvca-2020-term-sheet-2 |
| R18 | MIT Orbit, Term sheet pamphlet for founders | https://orbit-kb.mit.edu/hc/en-us/article_attachments/17392077450125 |

| Ref | Source | Link |
|---|---|---|
| R19 | VCBeast, VC Term Sheet Template and Guide | https://vcbeast.com/vc-term-sheet-template-guide |
| R20 | VCBeast, NVCA Term Sheet: The Standard Template | https://vcbeast.com/nvca-term-sheet-template |
| R21 | Vaquill, Liquidation Preference Math in Venture Deals (reporting Carta deal-terms data) | https://www.vaquill.ai/blog/liquidation-preference-math-venture-deals |
| R22 | Bird & Bird, How does a liquidation preference work | https://www.twobirds.com/en/insights/2023/global/how-does-a-liquidation-preference-work |
| R23 | Shoosmiths, Liquidation preferences | https://www.shoosmiths.co.uk/-/media/download-documents/support-for-founders/liquidation-preferences.pdf |
| R24 | Mondaq India, Liquidation Preference: A Downside Protection | https://www.mondaq.com/india/shareholders/1372484/liquidation-preference-a-downside-protection |
| R25 | VCBeast, Liquidation Preference Mechanics | https://vcbeast.com/liquidation-preference-mechanics-guide |
| R26 | Allied VC, Modelling Liquidation Preferences in Cap Tables | https://www.allied.vc/guides/modelling-liquidation-preferences-in-cap-tables |
| R27 | Altss, Anti-Dilution Protection glossary | https://altss.com/glossary/anti-dilution |
| R28 | Qapita, Anti-dilution provisions | https://www.qapita.com/blog/anti-dilution-provisions |
| R29 | Ryan O'Connell, Antidilution Provisions: Full Ratchet vs Weighted Average | https://ryanoconnellfinance.com/antidilution-provisions/ |
| R30 | Sutton, Understanding Anti-Dilution | https://sutton.substack.com/p/understanding-anti-dilution |
| R31 | Sequoia Capital, Writing a Business Plan | https://www.sequoiacap.com/article/writing-a-business-plan/ |
| R32 | Visible.vc, Investment Memo Template | https://visible.vc/blog/investment-memo-template/ |
| R33 | VCBeast, How to Write an Investment Memo (VC Template and Examples) | https://vcbeast.com/how-to-write-investment-memo-vc-template |

| Ref | Source | Link |
|---|---|---|
| R34 | VCBeast, How to Write an Investment Memo: Template and Examples (IC memo structure) | https://vcbeast.com/guides/how-to-write-an-investment-memo |
| R35 | Venture Capital Careers, Venture Capital Investment Memo | https://venturecapitalcareers.com/blog/venture-capital-investment-memo |
| R36 | Hustle Fund, Investment Memo Template | https://www.hustlefund.vc/post/investment-memo-template |
| R37 | VCBeast, VC Case Study Guide | https://vcbeast.com/vc-case-study-guide |
| R38 | Legal500 (Sarthak Advocates), Abolition of Angel Tax in India | https://www.legal500.com/developments/thought-leadership/abolition-of-angel-tax-in-india-a-boost-for-the-startup-ecosystem/ |
| R39 | Patron Accounting, Angel Tax Abolished: What Startups Must Know in 2026 | https://www.patronaccounting.com/blog/angel-tax-abolished-section-56-2-viib-startups-2026 |
| R40 | Brad Feld and Jason Mendelson, Venture Deals | https://venturedeals.com |
| R41 | Module 2 course material, Days 4 to 6: startup ecosystem, due diligence and the Sequoia due-diligence case study (MedAI Diagnostics and QuickBasket) | Course document |

# APPENDIX A | STUDENT ASSIGNMENT

## ROUTEWISE TECHNOLOGIES — INVESTMENT DECISION CASE

Purpose: apply the Module 3 curriculum to a realistic venture-financing decision. Choose ONE track: the Core Track tests the mandatory material without advanced calculations; the Advanced Track includes every Core requirement plus the optional advanced extensions.

| CHOICE | SCOPE | SUBMISSION PACKAGE |
|---|---|---|
| **TRACK A — CORE ONLY** | Mandatory topics from Days 7–9. No SAFE cap/discount conversion maths, option-pool sensitivity, valuation-method extensions, exit-waterfall calculations or anti-dilution formula work. | 1–2 page investment memo + calculation sheet + annotated term sheet + source log. |
| **TRACK B — CORE + ADVANCED** | Complete all Core requirements, then solve the advanced conversion, option-pool, valuation, exit-waterfall, anti-dilution and portfolio-monitoring extensions. | The same Core package + an advanced calculations appendix + a one-page portfolio dashboard. |

### 1. Case brief

RouteWise Technologies Pvt. Ltd. is a fictional Indian B2B software company. Its platform helps mid-sized distributors plan delivery routes, reconcile proof of delivery and monitor dispatch costs. Management is raising capital to expand sales coverage, strengthen the product and extend runway. Your role is to prepare an independent investment recommendation for a venture-capital investment committee.

#### Company snapshot: management-provided information

| METRIC / FACT | CASE INPUT | ANALYST NOTE |
|---|---|---|
| **Paying customers** | 120 | Request cohort, concentration and reference data. |
| **Annual recurring revenue (ARR)** | ₹2.4 crore | Treat this as a management-reported figure until verified. |
| **Year-on-year ARR growth** | 95% | Check the period, starting base and quality of growth. |
| **Gross margin** | 72% | Ask how implementation and support costs are classified. |
| **Annual logo churn** | 8% | Request cohort-level retention and a clear definition. |
| **Customer acquisition cost (CAC)** | ₹1.8 lakh | Ask whether sales salaries, commissions and marketing are included. |
| **Cash balance / monthly net burn** | ₹1.2 crore / ₹18 lakh | Calculate indicative runway; state limitations. |
| **Top 10 customers** | 38% of ARR | Assess concentration risk and renewal exposure. |
| **Management market claim** | TAM ₹8,000 crore; SAM ₹1,200 crore; five-year SOM ₹120 crore | Validate or challenge the assumptions with transparent evidence. |

### Proposed financing and capitalization

| ITEM | BASE-CASE ASSUMPTION |
|---|---|
| **Proposed raise** | ₹4 crore of new priced equity. |
| **Pre-money / post-money valuation** | ₹16 crore / ₹20 crore, before any separately modelled option-pool top-up. |
| **Existing fully diluted capitalization** | 8,000,000 founder ordinary shares + 500,000 existing ESOP shares/reserve = 8,500,000 shares. |
| **Proposed investor ownership** | 20% post-money in the base case, before separately modelled advanced pool sensitivity. |
| **Selected term-sheet points** | 1× non-participating liquidation preference; pro-rata right; investor board-observer right; reserved matters for a sale, new securities, large borrowings and related-party transactions; broad-based weighted-average anti-dilution, subject to negotiated exclusions. |
| **Future financing scenario** | For the Core cap-table exercise, assume a later investor receives 25% of the post-Round-2 company, with no other share issuances or pool changes. |

**Important:** RouteWise, all company metrics and all transaction terms above are fictional teaching inputs. Do not present them as claims about a real company. Clearly label management claims, independently verified facts, estimates and your own assumptions.

## 2. Track A — Core Topics Only

Complete every task below. The goal is a concise, evidence-led investment decision using the mandatory topics from Days 7, 8 and 9. You may describe advanced clauses in plain language, but you are not required to solve advanced conversion, option-pool, exit-waterfall or anti-dilution calculations.

### Task A1 — Choose the financing structure

In 250–350 words, compare priced equity, a convertible note and a SAFE. Recommend the best fit for this specific ₹4 crore round and explain the trade-offs. Discuss valuation certainty, documentation, conversion timing, investor rights and founder dilution. For the SAFE/note, explain the relevant concepts qualitatively; no cap/discount conversion calculation is required in this track.

### Task A2 — Valuation and cap table

1. Calculate the pre-money and post-money valuations and the proposed investor's post-money ownership under the stated base case.
2. Calculate the implied price per share and the number of new shares issued, using 8,500,000 existing fully diluted shares and the stated ₹16 crore pre-money valuation.
3. Prepare ownership tables for (i) before Round 1, (ii) after the proposed Round 1 and (iii) after the hypothetical Round 2. In Round 2, the new investor must own 25% post-round. Show shares and ownership percentages; reconcile the totals to approximately 100%.
4. Explain in plain language why the earlier holders' ownership percentages fall even though their share counts do not change in the simplified Round 2 scenario.

### Task A3 — Market assessment and diligence

- Build a transparent TAM/SAM/SOM view. Validate or challenge management's ₹8,000 crore / ₹1,200 crore / ₹120 crore market claims; show the logic and definitions behind each number.
- List five high-priority diligence questions. Cover customers and retention, unit economics, product differentiation, founder/team execution, and financial or legal risks.
- Use at least three credible, traceable sources. Cite the source next to the claim it supports and include the publisher, title, link and access date in your source log.

### Task A4 — Annotate the term sheet

Explain at least eight clauses in plain language and state one diligence question or negotiation point for each. Include the investment amount/security, pre- and post-money valuation, liquidation preference, pro-rata rights, board observer, reserved matters, ESOP treatment and anti-dilution. In the Core Track, explain the economics of a 1× non-participating preference conceptually; waterfall arithmetic is reserved for Track B.

### Task A5 — Investment committee memo

Write a 1–2 page memo that begins with one clear decision: INVEST, PASS or CONTINUE DILIGENCE. Include a deal header, company snapshot, two or three testable investment-thesis claims, market/competition, valuation and ownership, ranked risks with mitigants, and the next diligence actions that could change your recommendation. Avoid generic statements such as "large market" or "strong team" unless supported by evidence.

### Track A submission checklist

- 1–2 page investment memo with a clear recommendation.
- Calculation sheet showing valuation and cap tables across the two rounds.
- Annotated term sheet with at least eight clauses.
- Market-sizing note, five diligence questions and a source log.
- Explicit labels for assumptions, management claims and verified facts.

### Track A marking rubric — 100 points

| CRITERION | WEIGHT | WHAT STRONG WORK SHOWS |
|---|---|---|
| **Market understanding and evidence** | 20 | Transparent TAM/SAM/SOM; credible sources; clear customer segment and competition. |
| **Valuation and cap table** | 25 | Correct pre/post-money maths, share issuance, dilution and reconciled ownership percentages. |
| **Diligence quality** | 15 | Prioritised questions tied to decision-changing risks; facts separated from assumptions. |
| **Term-sheet understanding** | 15 | Accurate explanation of key economics, governance and ownership provisions. |
| **Investment thesis and recommendation** | 20 | Specific, testable thesis; ranked risks; reasoned invest/pass/continue-diligence decision. |

| CRITERION | WEIGHT | WHAT STRONG WORK SHOWS |
|---|---|---|
| **Clarity and source discipline** | 5 | Concise writing, traceable citations and consistent assumptions. |
| **TOTAL** | 100 | The memo must be useful to an investment committee, not merely descriptive. |

# 3. Track B — Core + Advanced Topics

Complete all Track A tasks, then add the extensions below. Track B tests the optional advanced material and requires calculations, assumptions and a brief interpretation of what each result means for the investment decision.

## Task B1 — SAFE cap and discount conversion

A separate hypothetical SAFE has an investment amount of ₹1 crore. Its cap price is ₹120 per share. The next priced-round share price is ₹160 and the SAFE gives a 20% discount to that price. Assume the contract uses the lower of the cap price and the discounted price; ignore interest, fees and other adjustments.

1. Calculate the discounted price and identify the conversion price that applies under the stated rule.
2. Calculate the number of shares issued on conversion and compare it with the result if only the discount applied.
3. Explain why the cap/discount result should be modelled using the instrument's actual contractual capitalization definition. Keep this a separate scenario unless you explicitly define how it enters RouteWise's base cap table.

## Task B2 — Option-pool top-up sensitivity

Compare the base case with a scenario in which the ESOP pool is increased to 10% of the post-money fully diluted capitalization immediately before the priced round, with the top-up borne by the pre-money holders. For this exercise, keep founder shares fixed at 8,000,000; the post-money investor stake fixed at 20%; and the post-money pool fixed at 10%. Treat pre-money valuation as ₹16 crore and the new investment as ₹4 crore.

1. Solve for total post-money shares, total pool shares, new investor shares and the additional pool shares required above the existing 500,000-share pool.
2. Calculate the implied price per share and compare founder, pool and new-investor ownership with the base case. State who bears the incremental dilution.
3. Explain why the option-pool size and its pre-/post-money treatment are negotiation variables and why actual legal definitions must control a live transaction model.

## Task B3 — Valuation cross-checks

Use the following illustrative inputs for the exercise only; they are not current market benchmarks or investment advice.

| METHOD | ILLUSTRATIVE INPUT | REQUIRED ANALYSIS |
|---|---|---|
| **Comparable-company multiple** | 6× ARR; RouteWise ARR = ₹2.4 crore; cash = ₹1.2 crore; debt = ₹0.2 crore. | Calculate implied enterprise value, then bridge to implied equity value. Compare it with the proposed ₹16 crore pre-money valuation and explain why a single multiple is not conclusive. |
| **VC method / return back-solve** | Exit equity value in five years = ₹100 crore; investor target = 10× MOIC; proposed cheque = ₹4 crore; assume 30% proportional ownership dilution before exit. | Calculate the ownership needed at exit and before later dilution to achieve the target. Then estimate the proposed investor's proceeds and MOIC if it starts at 20% post-money. |

## Task B4 — Exit waterfall sensitivity

Assume the new investor invested ₹4 crore for 20% ownership and holds a 1× liquidation preference. Ignore debt, taxes, transaction costs, other senior securities, participation caps other than the specified cap, and all other classes of equity. Use proceeds available to equity holders of ₹5 crore, ₹12 crore and ₹30 crore.

1. For each exit value, calculate the investor payout and common-holder residual under (i) 1× non-participating, (ii) 1× participating, and (iii) 1× participating capped at 2× total proceeds to the investor.
2. Explain where the structures create the largest difference and why the exit value, participation language, cap and conversion election matter.

## Task B5 — Anti-dilution and negotiation

Assume preferred shares were originally purchased at ₹10 per share. The company later raises ₹10 lakh at ₹5 per share. For the broad-based weighted-average illustration, use A = 1,000,000 pre-round fully diluted shares, B = new money divided by the original conversion price, and C = new money divided by the down-round price. Use the simplified formula: $CP_2 = CP_1 \times (A + B) / (A + C)$.

1. Calculate B, C and the adjusted conversion price under the weighted-average formula.
2. Compare the adjustment with full ratchet, which resets the conversion price to ₹5. For an illustrative original investment of ₹1 lakh, show the resulting share-equivalents under the original ₹10 price, full ratchet and weighted-average price.
3. Write one founder-side and one investor-side argument about the provision, then identify exclusions or negotiation points that should be checked in the definitive documents.

## Task B6 — Portfolio-monitoring dashboard

Prepare a one-page quarterly dashboard using the RouteWise case facts. Include ARR and growth, gross margin, churn, CAC, cash, monthly burn, indicative runway, customer concentration, three leading risks and the next action for each. Distinguish reported values from calculated metrics and forecasts; explain which additional data would be needed before a real investment decision.

## Track B submission package

- All Track A deliverables.
- Advanced calculation appendix covering SAFE conversion, option-pool sensitivity, both valuation methods, exit waterfalls and anti-dilution.
- One-page portfolio-monitoring dashboard.
- A short paragraph explaining which advanced calculation most changes (or fails to change) the final recommendation.

## Track B marking — Core 70 + Advanced 30 points

| COMPONENT | POINTS | EVIDENCE REQUIRED |
|---|---|---|
| **Core assignment** | 70 | Core analysis, market evidence, valuation/cap table, term-sheet literacy, diligence and investment memo. |
| **SAFE conversion + pool sensitivity** | 8 | Correct prices/share counts; assumptions and dilution allocation are explicit. |

| COMPONENT | POINTS | EVIDENCE REQUIRED |
|---|---|---|
| **Comparable + VC-method valuation** | 6 | Correct EV-to-equity bridge, return back-solve and interpretation. |
| **Exit waterfalls** | 6 | Correct payouts and residuals for all exits and all three structures. |
| **Anti-dilution + negotiation** | 5 | Correct formula application; economically accurate comparison and balanced negotiation rationale. |
| **Portfolio dashboard** | 5 | Useful monitoring metrics, transparent calculations and actionable risk triggers. |
| **TOTAL** | 100 | The advanced appendix must reconcile internally and use clearly stated assumptions. |

## 4. Submission and quality standards

- Format: submit the memo as PDF or DOCX; submit calculations in a clearly labelled spreadsheet or appendix. Use ₹ crore / ₹ lakh consistently and label units in every table.
- Sources: cite external facts close to the relevant claim. Include source title, publisher, direct link and access date. Never present the fictional case facts as real-world data.
- Model discipline: show formulas, assumptions, capitalization denominator and rounding. If a result depends on a legal definition, identify the definition that must be checked.
- Decision quality: state what additional evidence would change your recommendation. A well-supported PASS or CONTINUE DILIGENCE can score as highly as an INVEST recommendation.
- Integrity: complete your analysis independently, disclose any collaboration allowed by the facilitator, and do not fabricate market evidence or customer interviews.

### Final self-check before submission

| CHECK | YES / NO |
|---|---|
| **Does the first paragraph state the requested decision and why it matters now?** | ☐ Yes ☐ No |
| **Can every important figure be traced to a source, case input or explicit assumption?** | ☐ Yes ☐ No |
| **Do share counts and ownership percentages reconcile across each scenario?** | ☐ Yes ☐ No |
| **Are the three most important risks linked to specific diligence tests?** | ☐ Yes ☐ No |
| **Does the conclusion state what evidence would change the recommendation?** | ☐ Yes ☐ No |